

Define CS, fit into CC

1. CUSTOMER SEGMENT(S)

CS

6. CUSTOMER CONSTRAINTS

CC

5. AVAILABLE SOLUTIONS

AS

Explore AS, differentiate

Focus on J&P, tap into BE, understand RC

2. JOBS-TO-BE-DONE / PROBLEMS

J&P

What are the issues affecting the customers?

- Blood smear images are time consuming to be examined manually.
- This diagnosis depends on the experienced pathologists, which is not necessarily present.
- Great likelihood of misinterpretation in case of numerous samples.
- Delayed diagnosis poses a problem of late treatment of malaria patients.
- Primary healthcare facilities do not have sophisticated support systems of diagnoses.

The customers require a rapid, effective, and computerized way of diagnosing blood smear with malaria.

9. PROBLEM ROOT CAUSE

RC

What will the proposed solution do about the problem?

The suggested system offers a malaria detector, based on AI that:

- Smear blood images are automatically analyzed.
- Makes quick and precise predictions of malaria.
- Lessens the burden of doctors and lab technicians.
- Aids in diagnosis in regions where there are few medical professionals.
- Facilitates prevention and intervention of malaria.

It is a machine learning and medical imaging solution that enhances the efficiency of healthcare.

7. BEHAVIOUR

BE

What are the current behaviors of the customers?

- Take blood specimens of patients.
- Axenic blood preparations.
- Observing slides under microscopes.
- Diagnosis should be done by consulting expert pathologists.
- Go on to treatment with diagnostic reports.

All these measures are time-consuming and require human skills.

Focus on J&P, tap into BE, understand RC

3. TRIGGERS

TR

What causes customers to make a move?

- Malaria outbreak in some areas.
- Late reception of laboratory test results.
- Hospitals that are interested in enhancing the diagnostic efficiency.
- Increasing the use of Artificial Intelligence in healthcare.
- Requirement of more rapid screening in case of disease outbreaks.

Such stimuli make healthcare providers seek automated diagnostic solutions.

10. YOUR SOLUTION

SL

The solution that is offered is a malaria detection system based on AI that analyses an image of the microscopic blood smear with the help of the deep learning model.

Working of the System

- A blood smear image is uploaded by the user using a web interface.
- Analysis is done on the image via preprocessing.
- The image is inputted into a trained CNN model.
- The image is divided into the categories of the model:

8. CHANNELS of BEHAVIOUR

CH

8.1 ONLINE

- Send blood smear photograph to an online diagnostic platform.
- Fraudulently access hospital computer systems.
- Digitally monitor prediction team results.

8.2 OFFLINE

- Gathering of blood samples of patients.

Identify strong TR & EM	4. EMOTIONS: BEFORE / AFTER EM The attitude of the customers before and after the resolution of the problem. Before - Physicians are under pressure because of work overload. - Patients worry and are unsure of the results of diagnoses. - The technicians are under pressure when they are analyzing a large number of samples manually. After - Physicians are comfortable with AI-assisted diagnosis. - The lab technicians are relieved by having lesser workload. - Patients feel secure and confident with expedited diagnosis.	Being parasitized (Malaria infected) Uninfected The prediction result is shown in the system with the confidence score. Benefits - Faster malaria diagnosis. - Reduced manual workload. - Better diagnostic error. - Hospitals, laboratories and rural health care facilities.	- Creation of microscope slides in labs. - Image like capturing of microscopes with imaging equipment. - Physicians who rely on outcomes to make a diagnosis.	Identify strong TR & EM